

Experiment 6: Breadth First Search (BFS)

Aim:

To implement Breadth First Search (BFS) for traversing a graph and demonstrate level-by-level state-space search.

Algorithm

1. Start
2. Initialize the graph and starting vertex.
3. Create a queue and mark the starting vertex as visited.
4. Insert the starting vertex into the queue.
5. Remove a vertex from the front of the queue and display it.
6. Add all unvisited adjacent vertices to the queue and mark them visited.
7. Repeat steps 5–6 until the queue is empty.
8. Stop.

Flowchart

Start



Initialize



Process / Search



Goal / Condition Reached?

n n

Yes No



Print Result Repeat



Stop

Objective

- Understand breadth-first state-space search.
- Traverse a graph systematically level by level.
- Use a queue for implementing BFS.

Program

```

from collections import deque

def bfs(graph, start):
    visited = set()
    queue = deque([start])
    visited.add(start)
    while queue:
        vertex = queue.popleft()
        print(vertex, end=" ")
        for neighbour in graph.get(vertex, []):
            if neighbour not in visited:
                visited.add(neighbour)
                queue.append(neighbour)

graph = {
    'A': ['B', 'C'],
    'B': ['D', 'E'],
    'C': ['F'],
    'D': [],
    'E': [],
    'F': []
}

print("BFS Traversal:")
bfs(graph, 'A')

```

Output

BFS Traversal:

A B C D E F

Result

The program was executed successfully and produced the expected result.

Conclusion

The experiment successfully demonstrated the corresponding Artificial Intelligence technique and its application